

Heat treatment of steel

19-205-0305:METALLURGY AND MATERIAL SCIENCE

THE IRON-IRON CARBIDE (Fe-Fe₃C) PHASE DIAGRAM

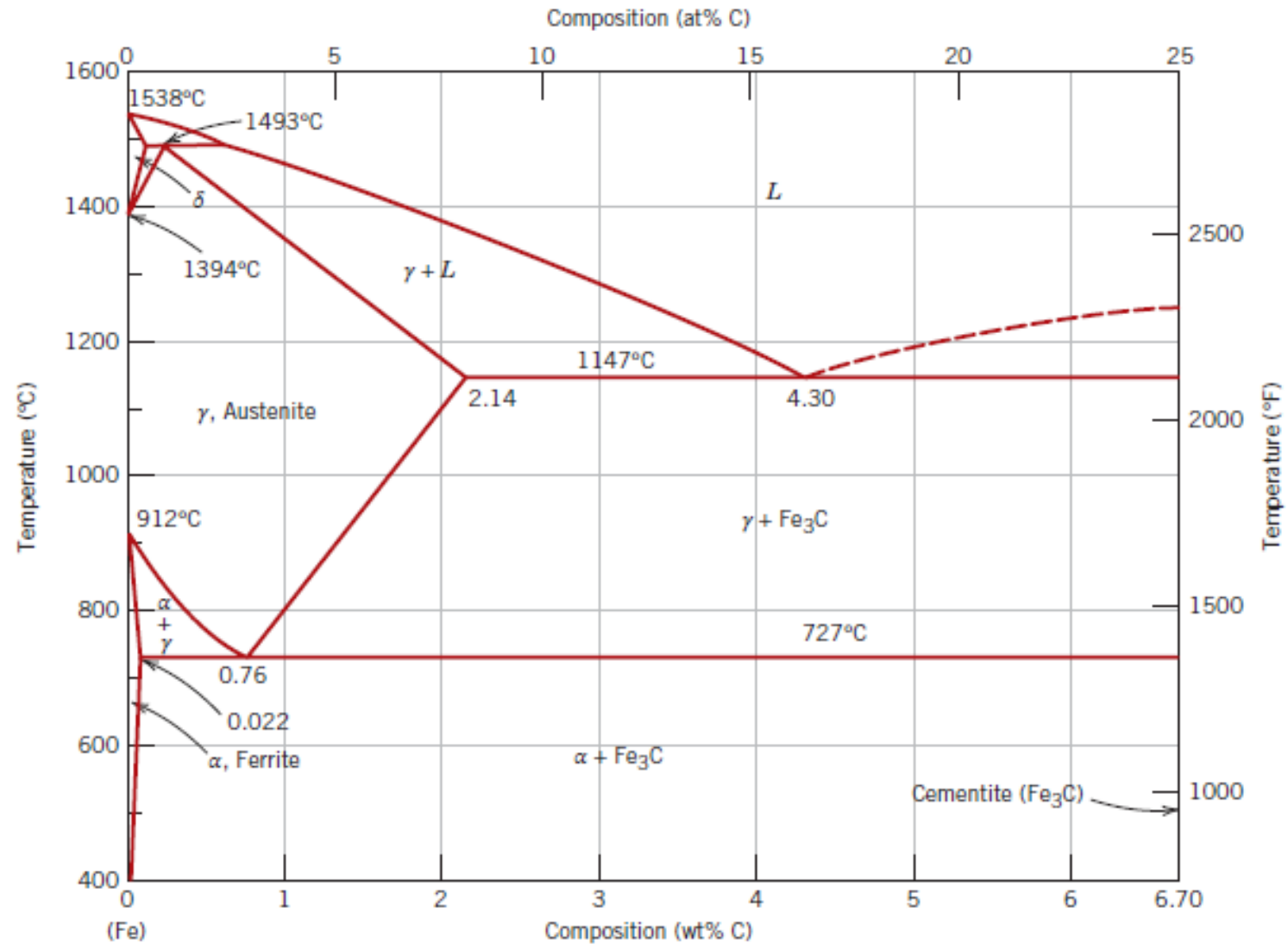
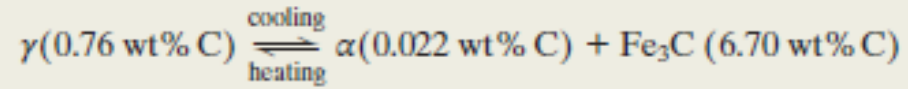
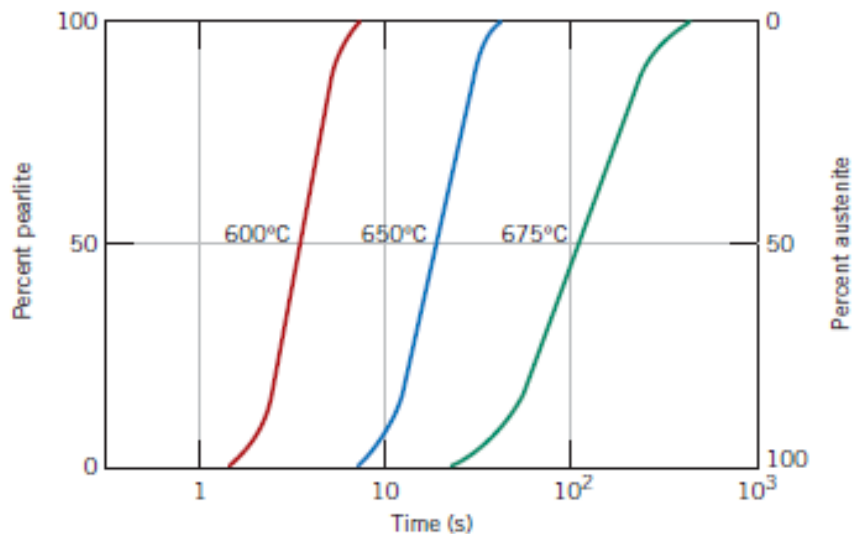


Figure 9.24 The iron-iron carbide phase diagram.

ISOTHERMAL TRANSFORMATION DIAGRAMS



Upon cooling, austenite, having an intermediate carbon concentration, transforms into a ferrite phase, which has a much lower carbon content, and also cementite, which has a much higher carbon concentration. Pearlite is one microstructural product of this transformation.



For an iron-carbon alloy of eutectoid composition (0.76 wt% C), the isothermal fraction reacted versus the **logarithm of time** for the **austenite-to-pearlite transformation**. For each curve, data were collected after rapidly cooling a specimen composed of 100% austenite to the temperature indicated; that temperature was maintained constant throughout the course of the reaction.

ISOTHERMAL TRANSFORMATION DIAGRAMS

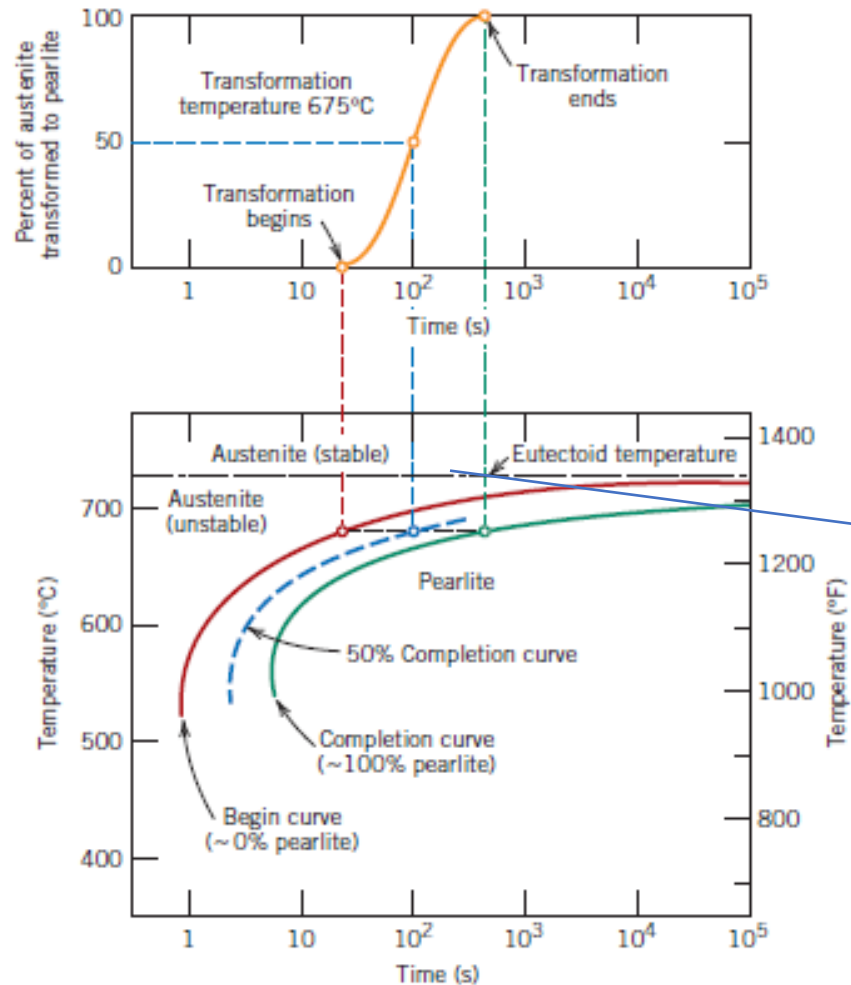


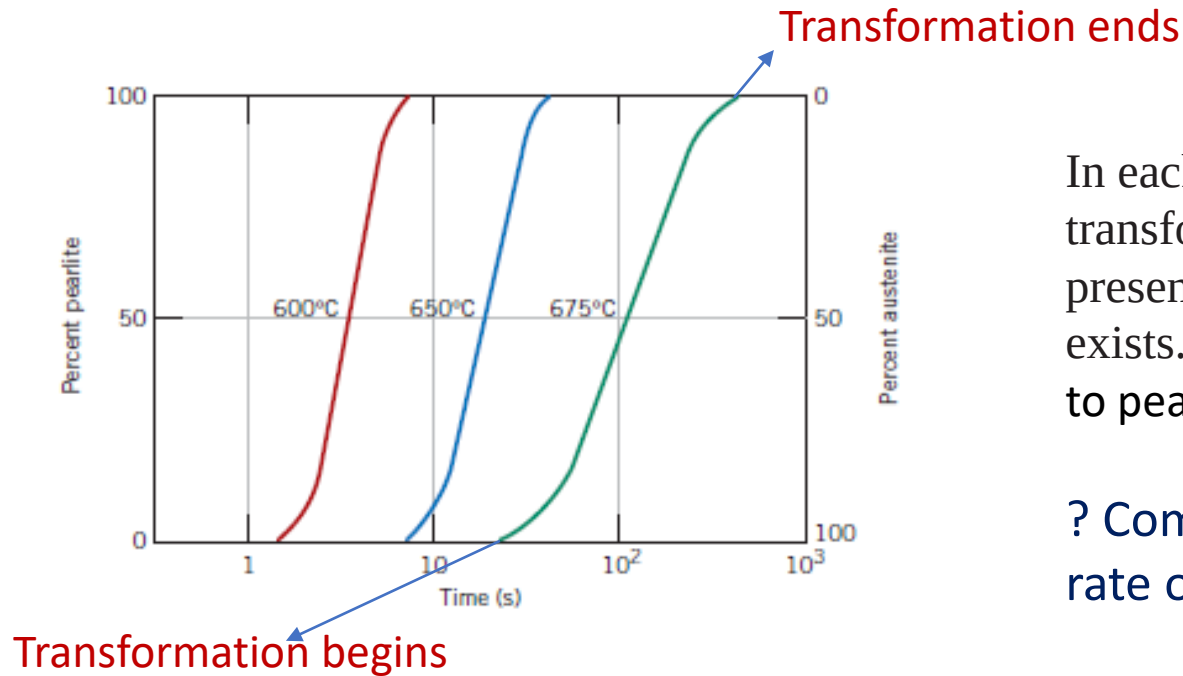
Figure: Demonstration of how an isothermal transformation diagram (bottom) is generated from percentage transformation versus- logarithm of time measurements (top).

Here, the vertical and horizontal axes are temperature and the logarithm of time. Two solid curves are plotted; one represents the time required at each temperature. for the initiation or start of the transformation, and the other is for the transformation conclusion.

The horizontal line indicates the eutectoid temperature (727°C); at temperatures above the eutectoid and for all times, only austenite exists, as indicated in the figure.

The austenite-to-pearlite transformation occurs only if an alloy is supercooled below the eutectoid; as indicated by the curves, the time necessary for the transformation to begin and then end depends on temperature.

ISOTHERMAL TRANSFORMATION DIAGRAMS



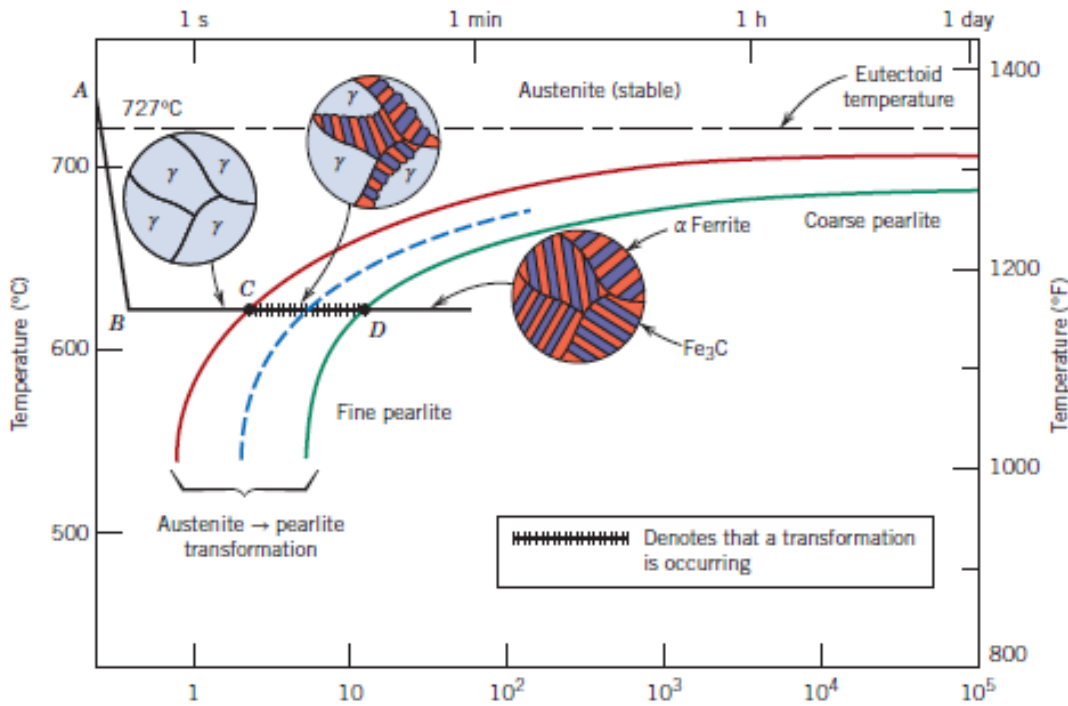
In each figure (at 600°C, 650°C and 675°C), on the left of the transformation start curve, only austenite (which is unstable) is present, whereas to the right of the finish curve, only pearlite exists. In between, the austenite is in the process of transforming to pearlite, and thus both microconstituents are present.

? Compare the transformation curves with regard to the rate of transformation and the isothermal temperature.

This particular plot is valid only for an iron-carbon alloy of eutectoid composition; for other compositions, the curves have different configurations.

Since the temperature is held constant during the transformation, this curve is called an **isothermal transformation diagram** or sometimes a **time-temperature-transformation (or T-T-T)** plot.

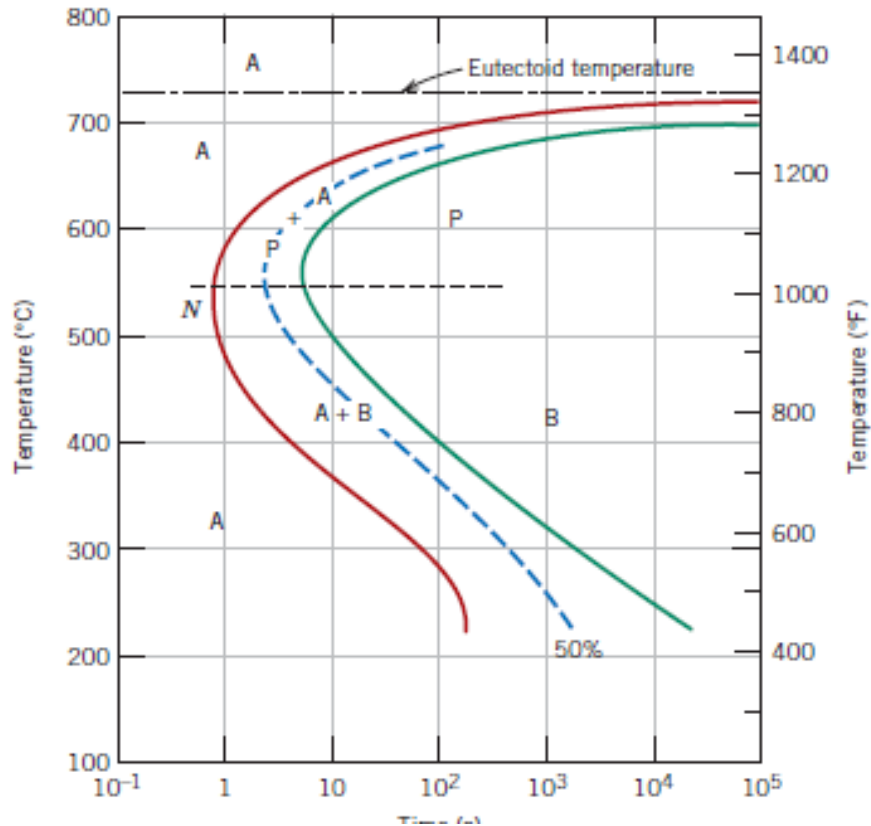
ISOTHERMAL TRANSFORMATION DIAGRAMS



An actual isothermal heat treatment curve (*ABCD*) is superimposed on the isothermal transformation diagram for a eutectoid iron-carbon alloy in the figure. Very rapid cooling of austenite to a given temperature is indicated by the near-vertical line *AB*, and the isothermal treatment at this temperature is represented by the horizontal segment *BCD*. Time increases from left to right along this line. The transformation of austenite to pearlite begins at point *C* (after approximately 3.5 s). It reaches completion in about 15 seconds, corresponding to point *D*. The figure also shows schematic microstructures at various times during the progression of the reaction.

? At higher isothermal temperatures coarse pearlite is formed; however, at lower temperatures fine pearlite is formed.

ISOTHERMAL TRANSFORMATION DIAGRAMS



Isothermal transformation diagram for an iron-carbon alloy of eutectoid composition, including austenite-to-pearlite (A–P) and austenite-to-bainite (A–B) transformations.

Bainite

The microstructure of bainite consists of ferrite and cementite phases, and thus, diffusional processes are involved in its formation. Bainite forms as needles or plates, depending on the temperature of the transformation; the microstructural details of bainite are so fine that their resolution is possible only using electron microscopy. Transformation to bainite occurs at temperatures below those at which pearlite forms.

The T-T-T curves are C-shaped and have a “nose” at point *N*, where the rate of transformation is a maximum. pearlite forms above the nose (i.e., over the temperature range of about 540°C to 727°C) at temperatures between about 215°C and 540°C bainite is the transformation product.

ISOTHERMAL TRANSFORMATION DIAGRAMS

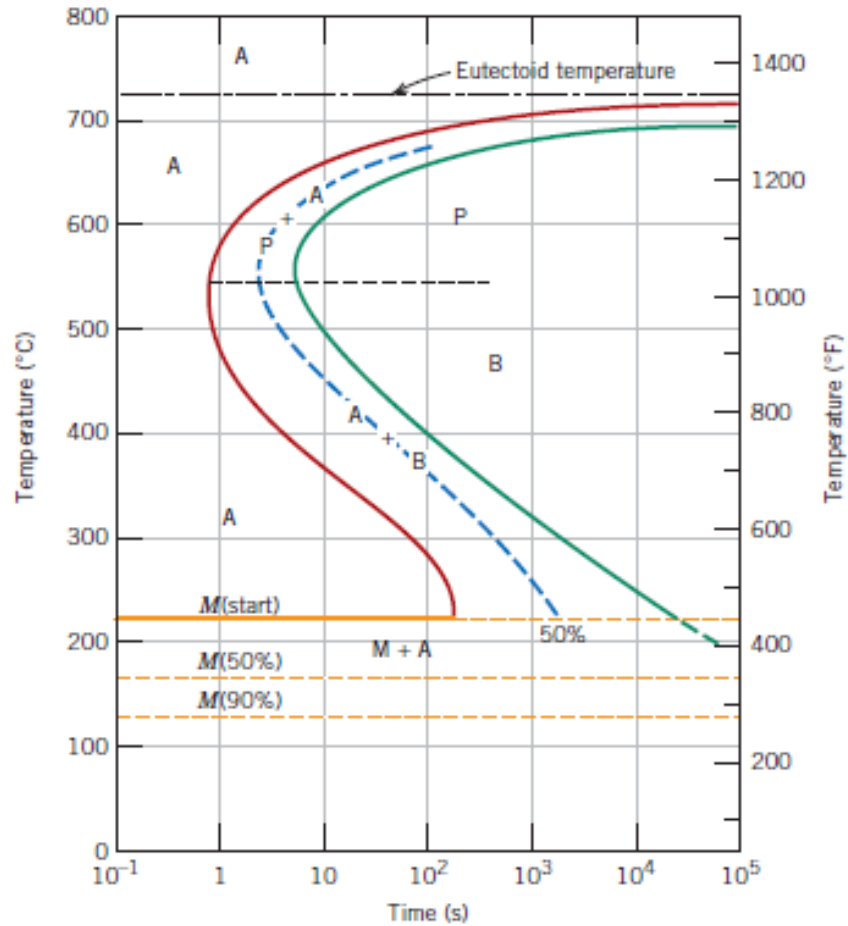
Spheroidite

If a steel alloy having either pearlitic or bainitic microstructures is heated to and left at a temperature below the eutectoid for a sufficiently long period of time, for example, at about 700°C for between 18 and 24 h yet another microstructure will form called **spheroidite**. Instead of the alternating ferrite and cementite lamellae (pearlite) or the microstructure observed for bainite, the Fe₃C phase appears as sphere-like particles embedded in a continuous α -phase matrix.

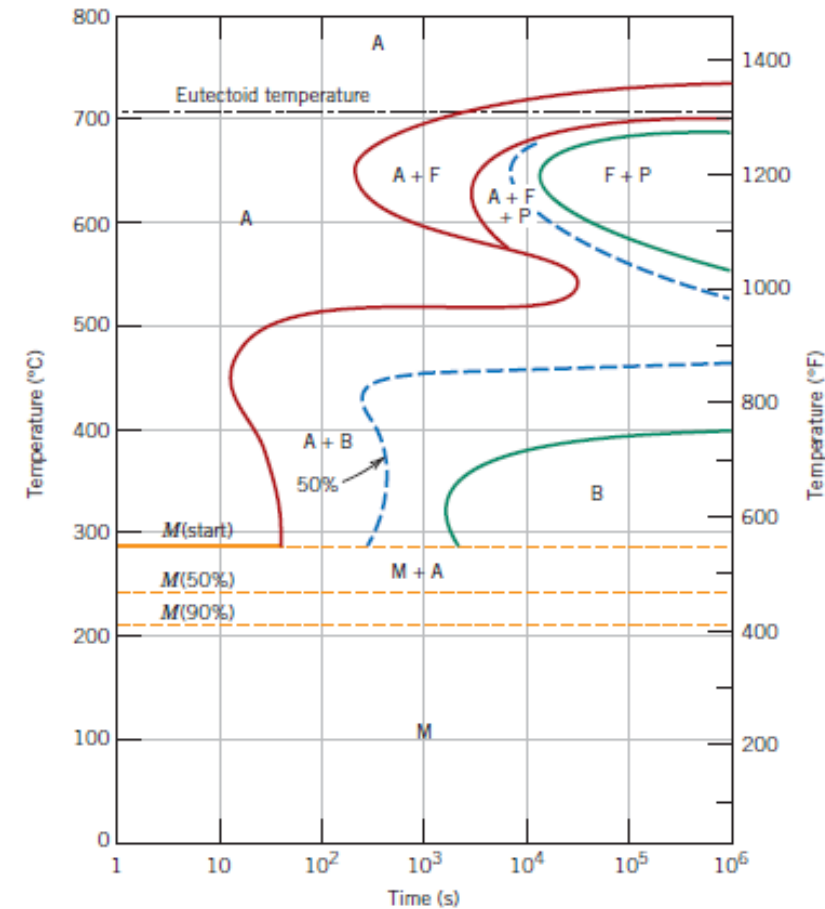
Martensite

Yet another microconstituent or phase called **martensite** is formed when austenitised iron-carbon alloys are rapidly cooled (or quenched) to a relatively low temperature (in the vicinity of the ambient). Martensite is a nonequilibrium single-phase structure that results from a diffusionless transformation of austenite. It may be considered a transformation product that is competitive with pearlite and bainite. The martensitic transformation occurs when the quenching rate is rapid enough to prevent carbon diffusion. Any diffusion whatsoever results in the formation of ferrite and cementite phases.

ISOTHERMAL TRANSFORMATION DIAGRAMS

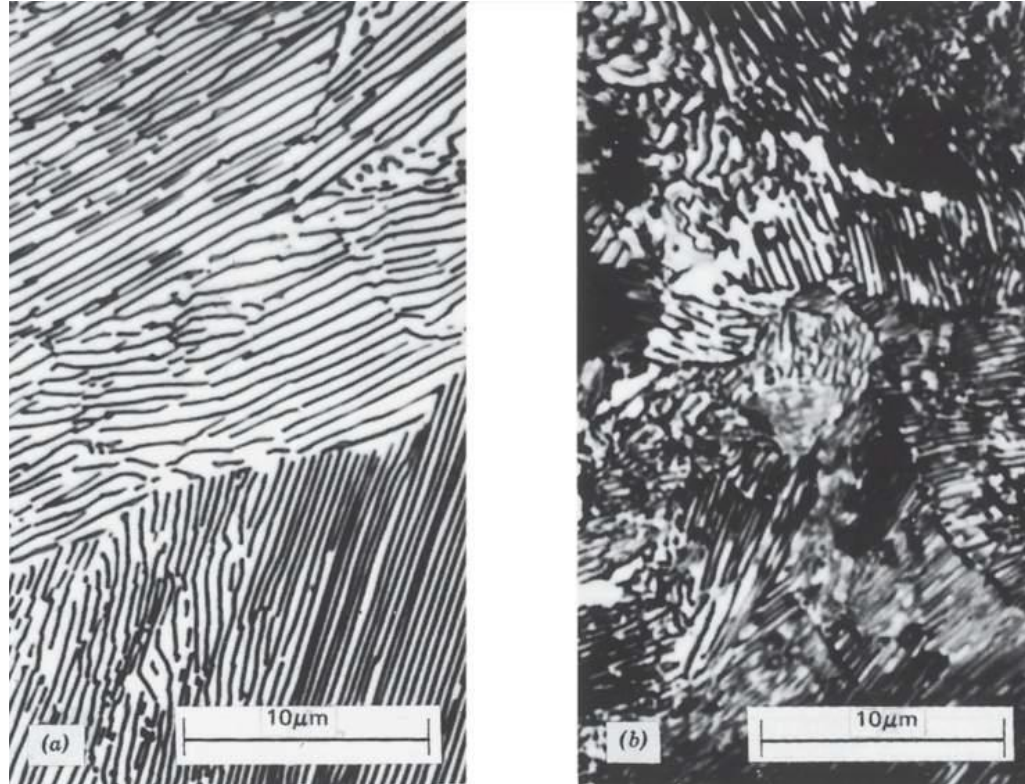


The complete isothermal transformation diagram for an iron-carbon alloy of eutectoid composition: A, austenite; B, bainite; M, martensite; P, pearlite.

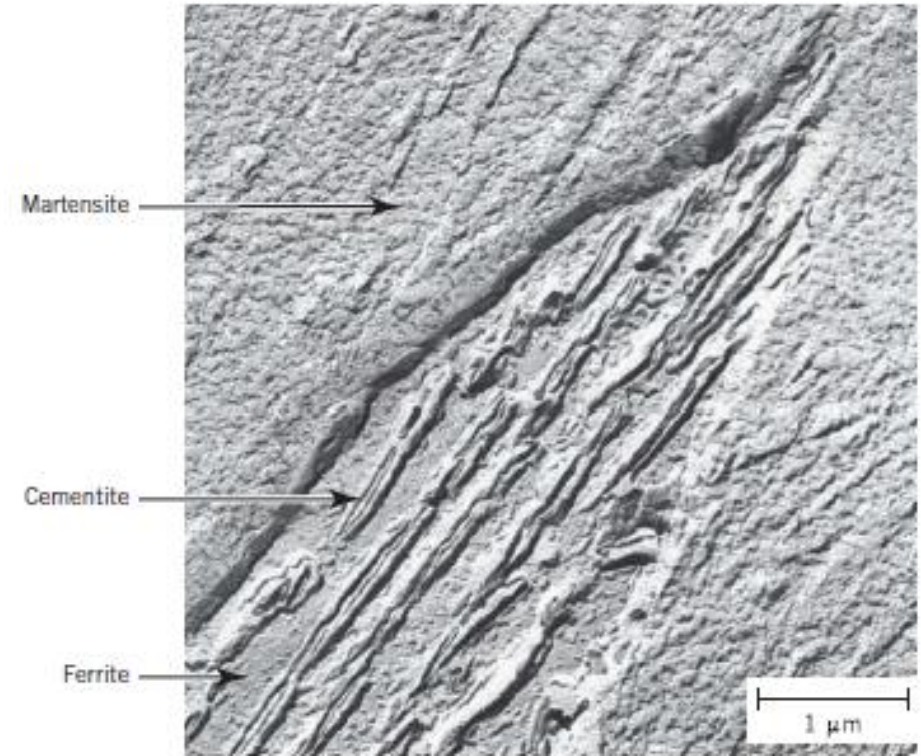


Isothermal transformation diagram for an alloy steel (type 4340): A, austenite; B, bainite; P, pearlite; M, martensite; F, pro eutectoid ferrite.

Photomicrographs of different microstructures

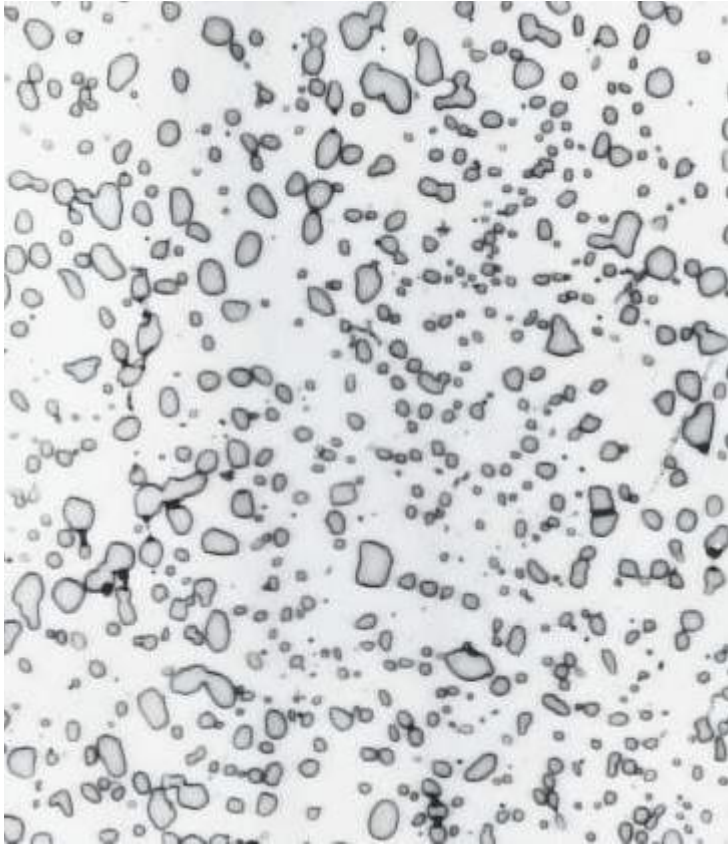


Photomicrographs of (a) coarse pearlite and (b) fine pearlite (3000×).



Transmission electron micrograph showing the structure of bainite. A grain of bainite passes from the lower left to upper right corners; it consists of elongated and needle-shaped particles of Fe_3C within a ferrite matrix. The phase surrounding the bainite is martensite.

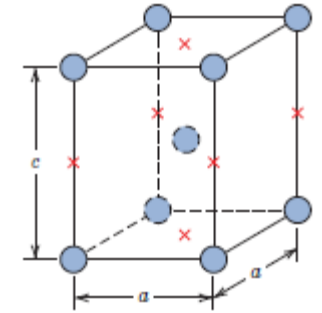
Photomicrographs of different microstructures



Photomicrograph of a steel having a spheroidite microstructure. The small particles are cementite and the continuous phase is α -ferrite (1000 $\times$).

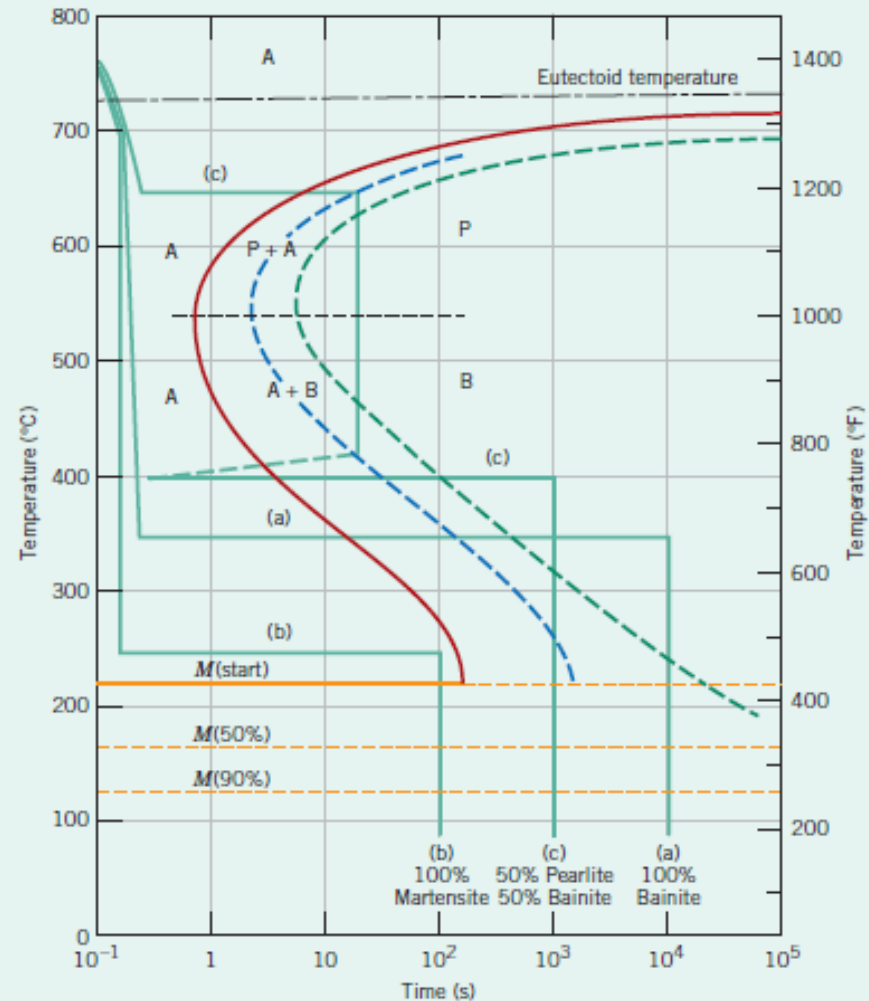


Photomicrograph showing the martensitic microstructure. The needle-shaped grains are the martensite phase, and the white regions are austenite that failed to transform during the rapid quench (1220 $\times$).



The body-centred tetragonal unit cell for martensitic steel shows iron atoms (circles) and sites that may be occupied by carbon atoms ($\times$ s). For this tetragonal unit cell, $c > a$.

Microstructural determinations for three isothermal heat treatments



Isothermal transformation diagram for the alloy steel (type 4340): A, austenite; B, bainite; P, pearlite; M, martensite; F, pro eutectoid ferrite.

ANNEALING PROCESSES

The term **annealing** refers to a heat treatment in which a material is exposed to an elevated temperature for an extended time period and then slowly cooled. Typically, annealing is carried out to (1) relieve stresses, (2) increase softness, ductility, and toughness, and/or (3) produce a specific microstructure. A variety of annealing heat treatments are possible.

Any annealing process consists of three stages: (1) **heating to the desired temperature**, (2) **holding or “soaking” at that temperature**, and (3) **cooling, usually to room temperature**.

Process Annealing

Process annealing is a heat treatment that is used to negate the effects of cold work that is, to soften and increase the ductility of a previously strain-hardened metal.

Stress Relief Annealing

Internal residual stresses may develop in metal pieces in response to the following: (1) plastic deformation processes such as machining and grinding; (2) nonuniform cooling of a piece that was processed or fabricated at an elevated temperature, such as a weld or a casting; and (3) a phase transformation that is induced upon cooling in which parent and product phases have different densities. Distortion and warpage may result if these residual stresses are not removed. They may be eliminated by a **stress relief** annealing heat treatment in which the piece is heated to the recommended temperature, held there long enough to attain a uniform temperature, and finally cooled to room temperature in air.

Annealing of Ferrous Alloys

The horizontal line at the eutectoid temperature, conventionally labelled A_1 , is termed the **lower critical temperature**, below which, under equilibrium conditions, all austenite has transformed into ferrite and cementite phases. The phase boundaries denoted as A_3 and A_{cm} represent the **upper critical temperature** lines for hypoeutectoid and hypereutectoid steels, respectively.

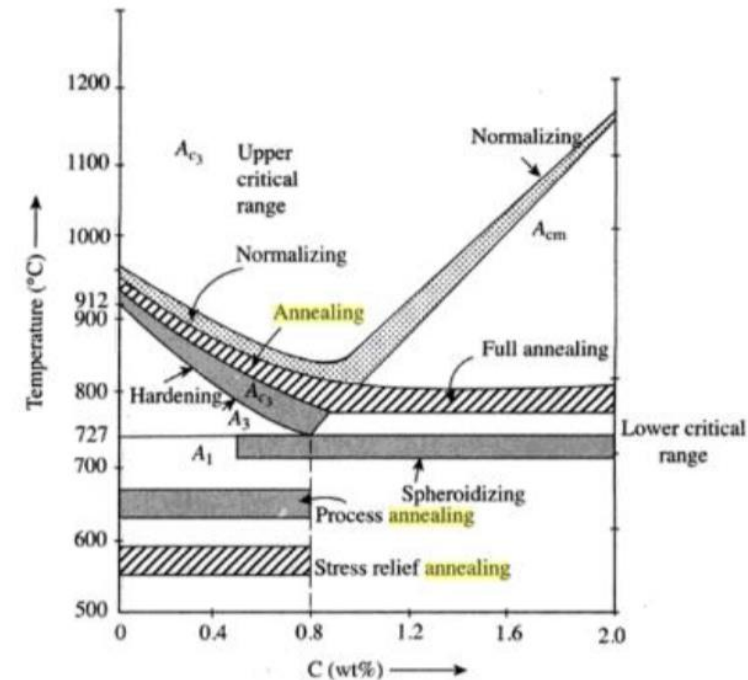
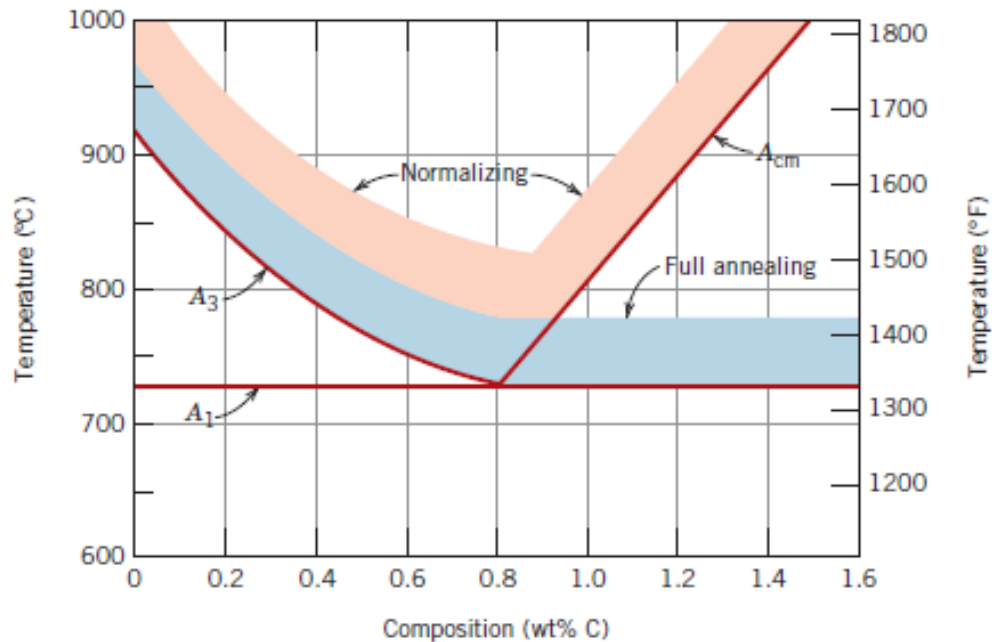
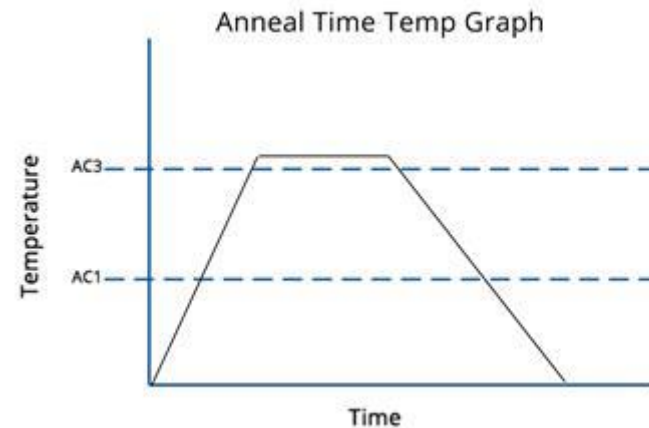
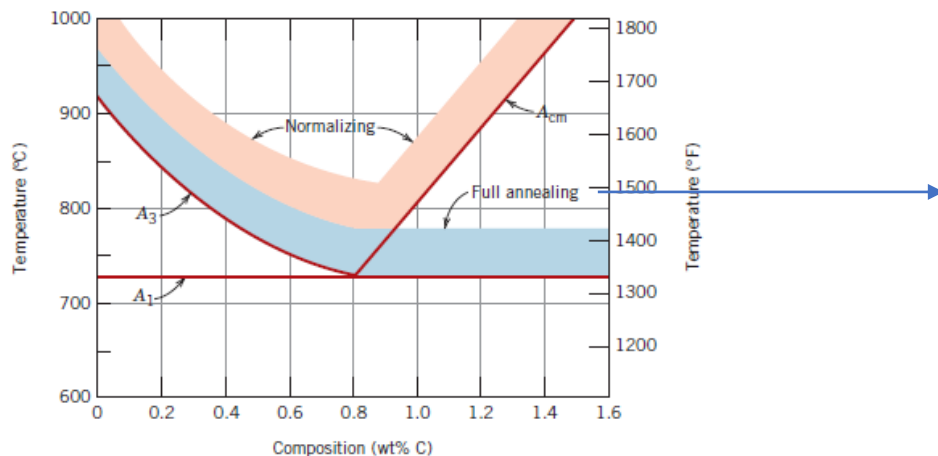


Fig. Heat Treatment Process

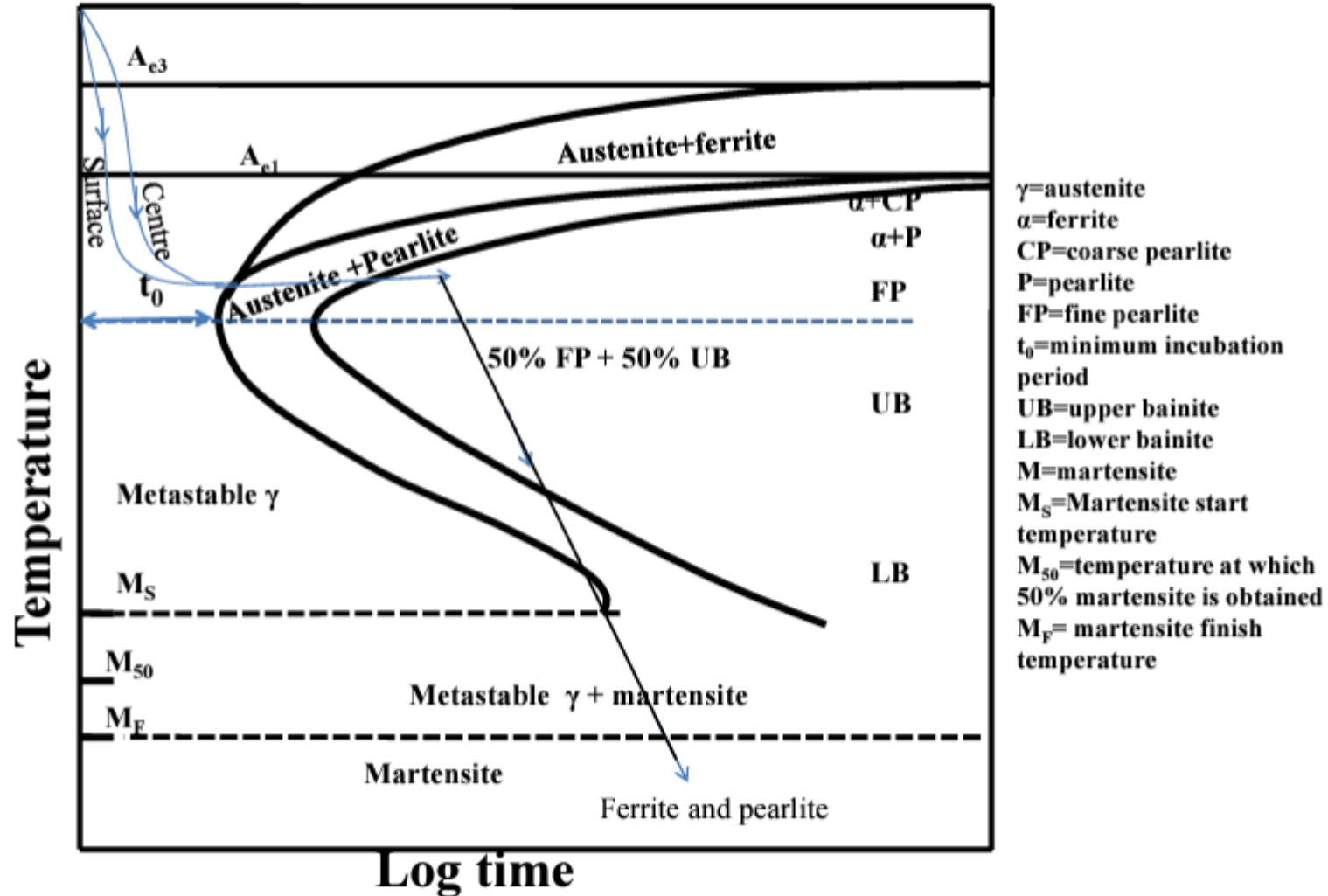
Annealing of Ferrous Alloys

Full Anneal

A heat treatment known as **full annealing** is often used in low- and medium-carbon steels that will be machined or will experience extensive plastic deformation during a forming operation. In general, the alloy is treated by heating to a temperature of about 50°C above the A_3 line (to form austenite) for compositions less than the eutectoid or, for compositions in excess of the eutectoid, 50°C above the A_1 line (to form austenite and Fe_3C phases), as noted in the Figure. The alloy is then furnace cooled, that is, the heat-treating furnace is turned off, and both furnace and steel cool to room temperature at the same rate, which takes several hours. The microstructural product of this anneal is coarse pearlite (in addition to any proeutectoid phase) that is relatively soft and ductile.



Isothermal annealing heat treatment superimposed on the TTT diagram of plain carbon hypoeutectoid steel



Normalising

Steels that have been plastically deformed by, for example, a rolling operation consist of grains of pearlite (and most likely a proeutectoid phase), which are irregularly shaped and relatively large and vary substantially in size. An annealing heat treatment called **normalising** is used to refine the grains (i.e., to decrease the average grain size) and produce a more uniform and desirable size distribution; fine-grained pearlitic steels are tougher than coarse-grained ones. Normalising is accomplished by heating at least 55°C above the upper critical temperature, that is, above A_3 for compositions less than the eutectoid (0.76 wt% C), and above A_{cm} for compositions greater than the eutectoid, as represented in the Figure.

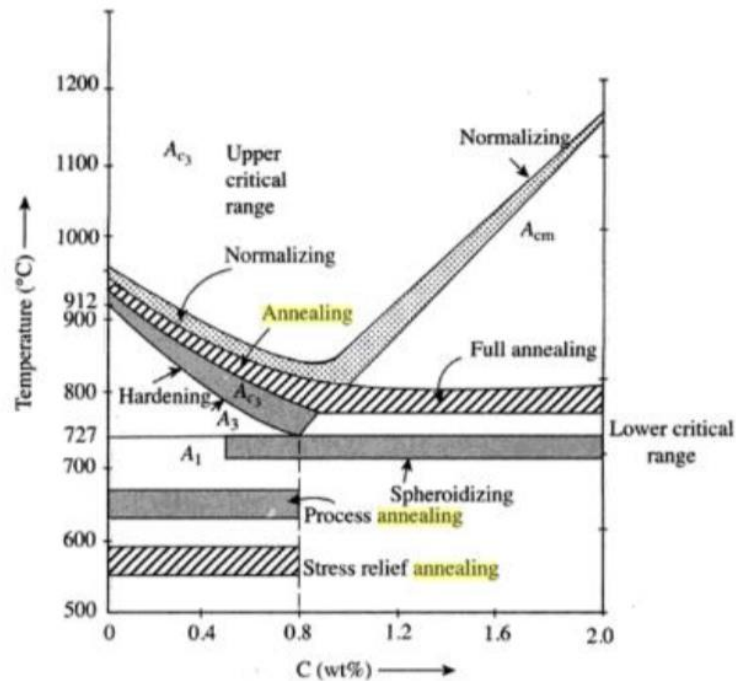


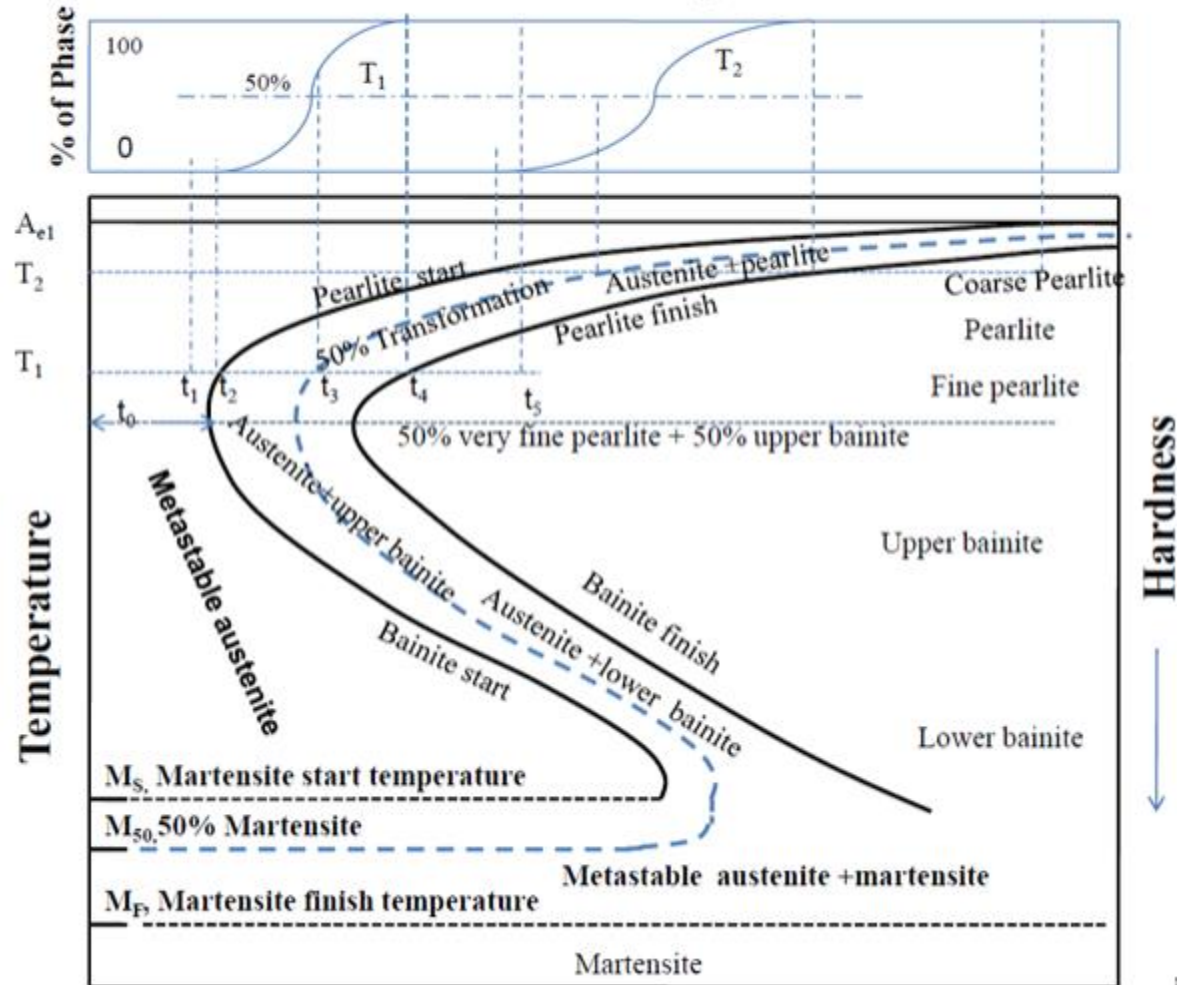
Fig. Heat Treatment Process

Spheroidizing

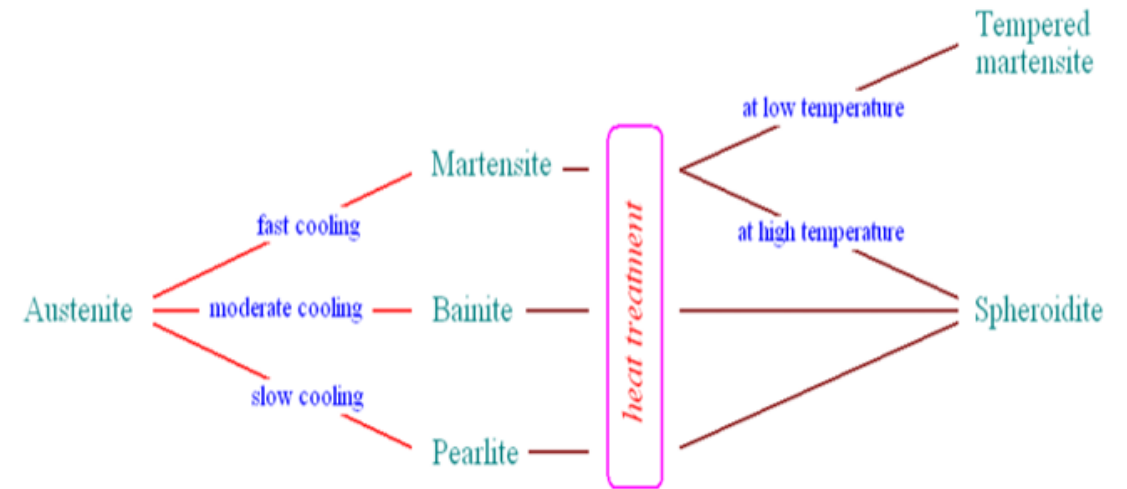
Medium- and high-carbon steels having a microstructure containing even coarse pearlite may still be too hard to machine or plastically deform conveniently. These steels, and in fact, any steel, may be heat-treated or annealed to develop the spheroidite structure as described in Section 10.5. Spheroidized steels have maximum softness and ductility and are easily machined or deformed. The **spheroidizing** heat treatment, during which there is a coalescence of the Fe_3C to form the spheroid particles.

Hardening

T-T-T Diagram



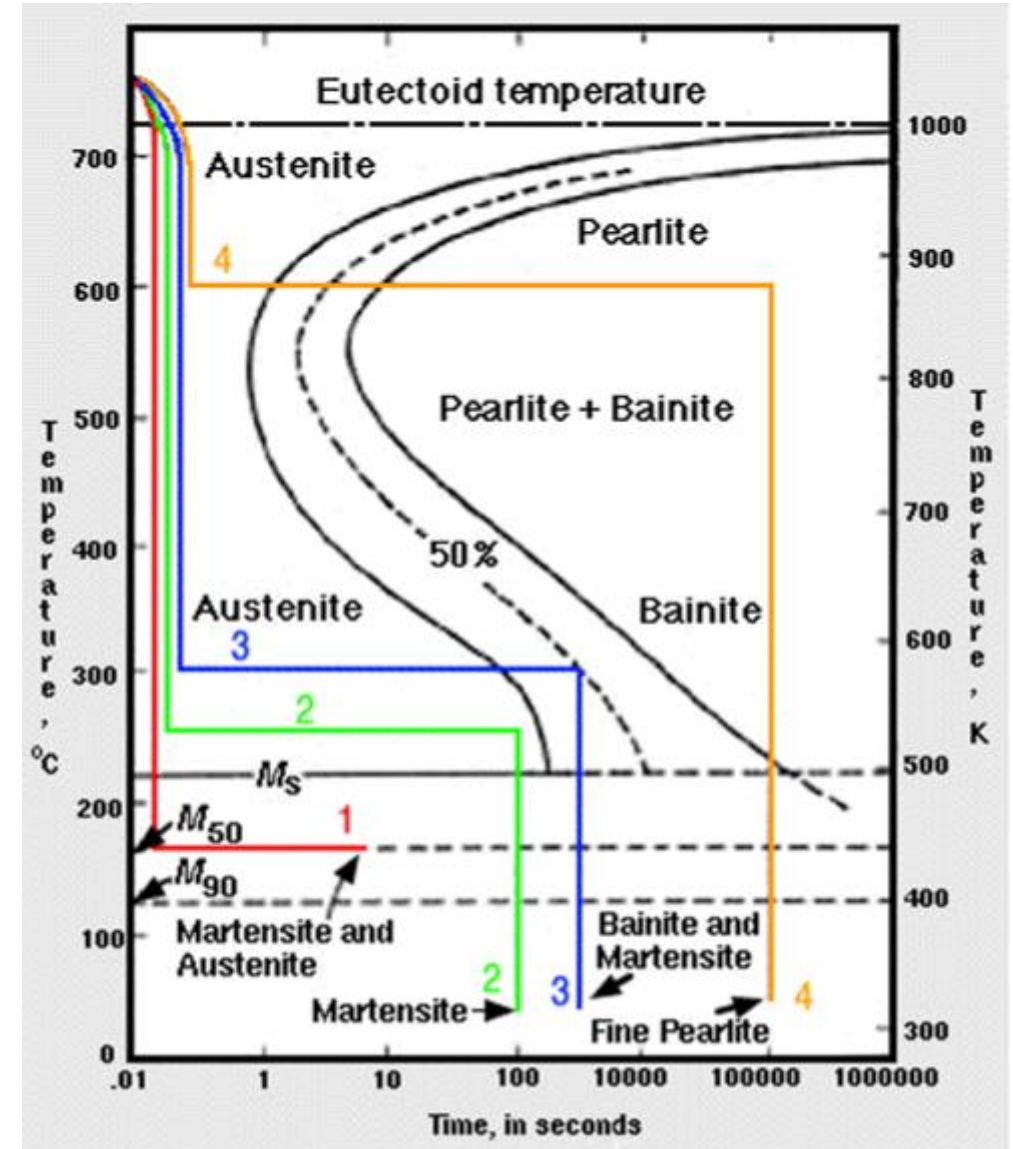
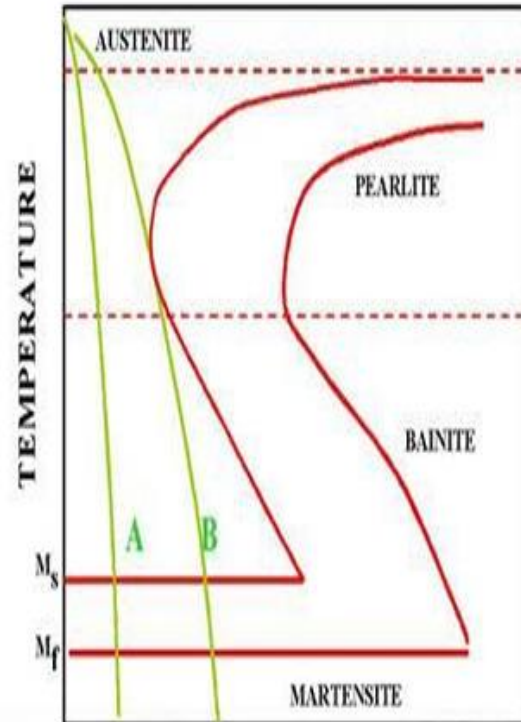
Transformations involving austenite for Fe-C system



Hardening- Quenching

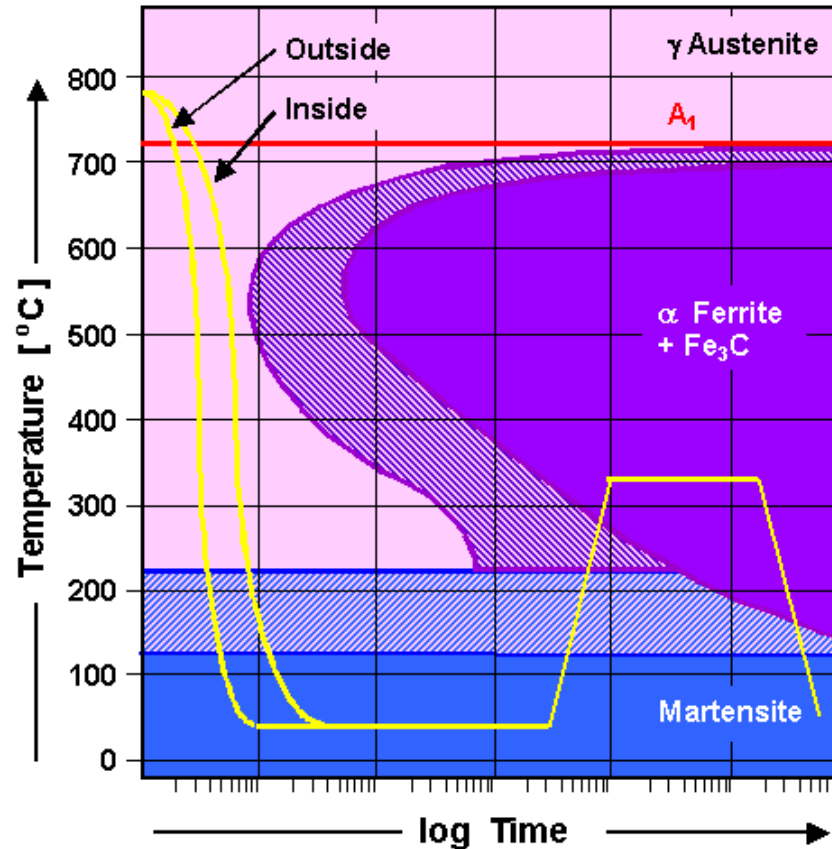
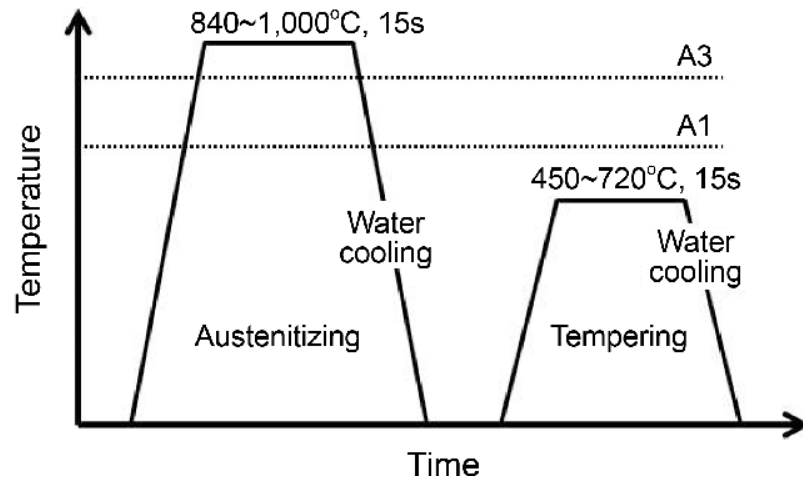
Quenching

- If cooling rate is very high, all Austenite will transform to Martensite
- Higher cooling rate (Rate A)
 - Higher distortion
 - Higher stresses than cooling rate B.
- The end product of both cooling rates is martensite.
- Cooling rate B is also known as the Critical Cooling Rate

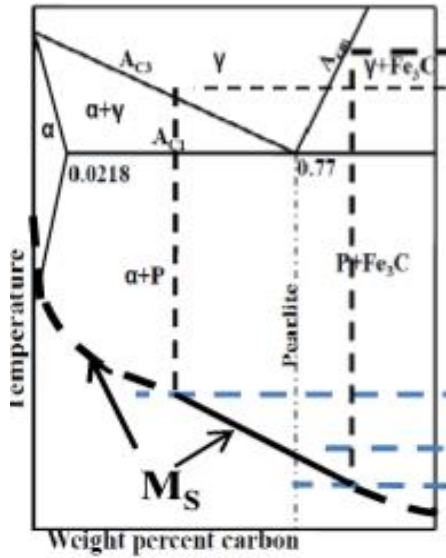


Hardening- Tempering

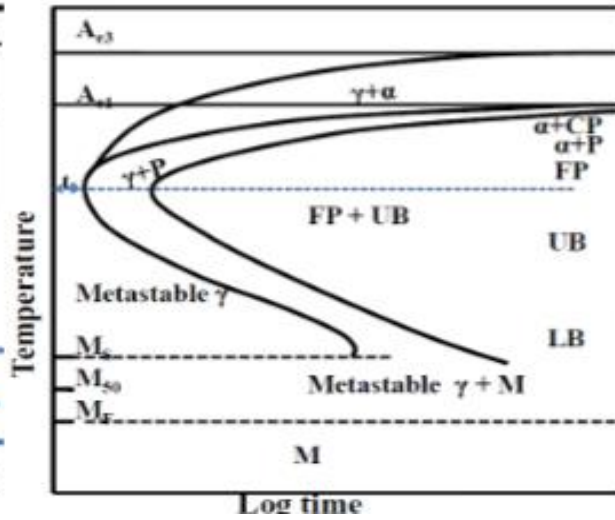
Tempering is a heat treatment process that improves the characteristics of a metal, such as steel, by heating it to a high temperature and then cooling it. The process can make the metal tougher by decreasing its hardness and brittleness and increasing its ductility.



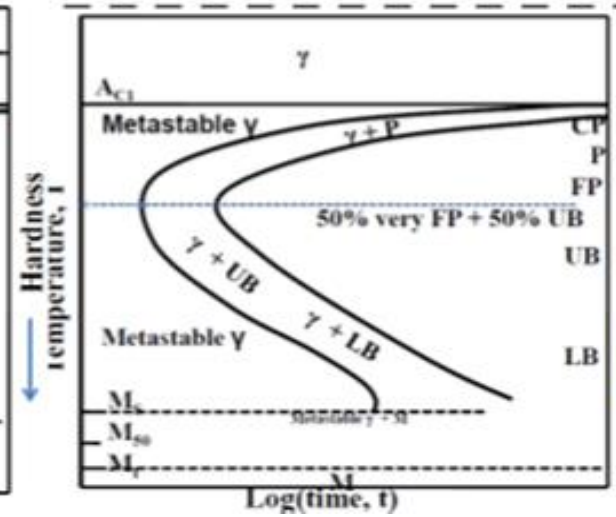
Hardening- Quenching



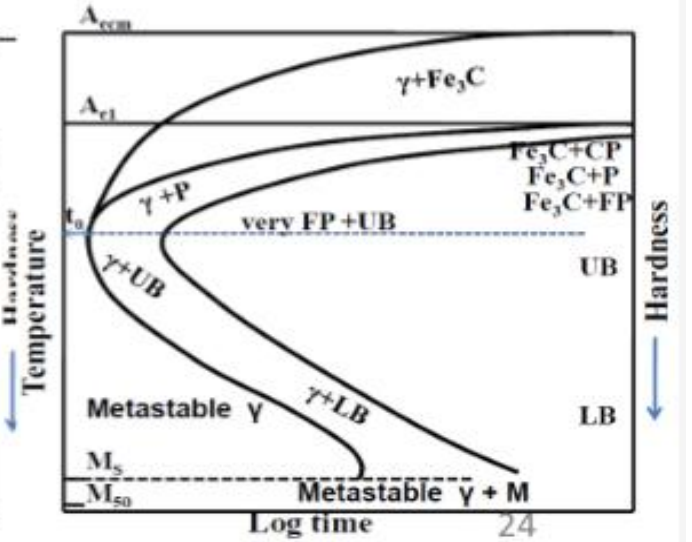
Fe – Fe₃C
Metastable
Phase diagram



TTT diagram for
hypoeutectoid steel

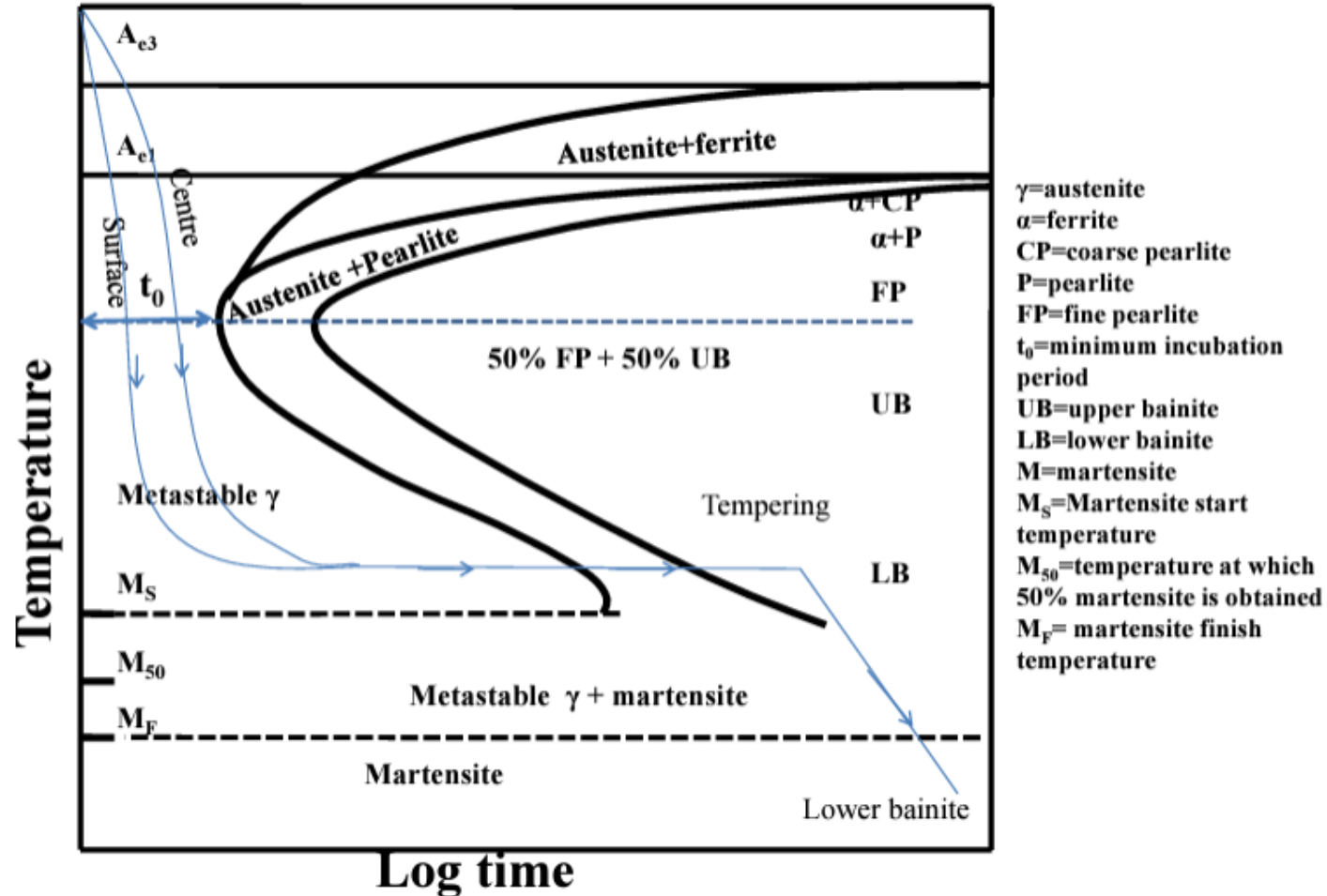


TTT diagram for
eutectoid steel



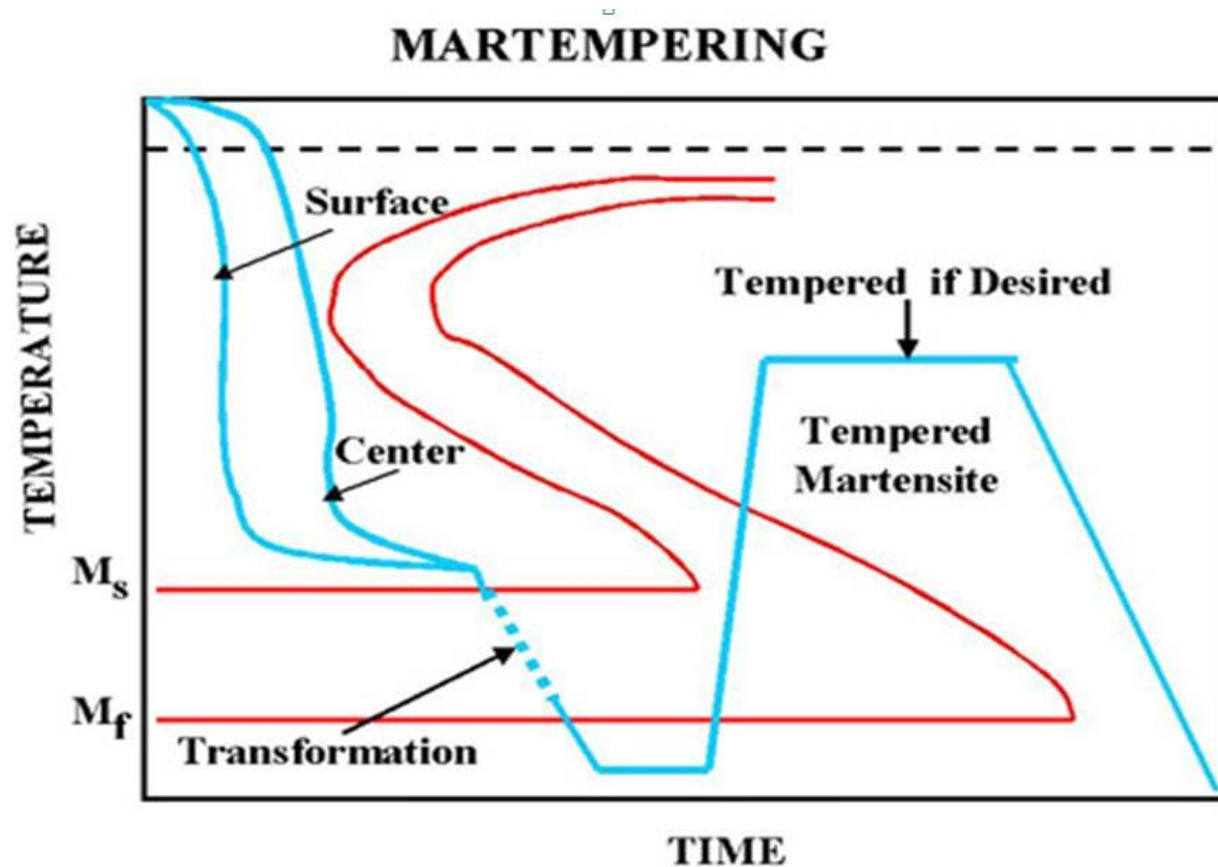
TTT diagram for
hypereutectoid steel

Austempering heat treatment superimposed on the TTT diagram for plain carbon hypoeutectoid steel



Austempering is a heat-treating process for medium-to-high carbon ferrous metals which produces a metallurgical structure called bainite. It is used to increase strength, and toughness and to reduce distortion. Parts are heated to the hardening temperature, then cooled rapidly enough to a temperature above the martensite start (M_s) temperature and held for a time sufficient to produce the desired bainite microstructure.

Martempering heat treatment superimposed on the TTT diagram



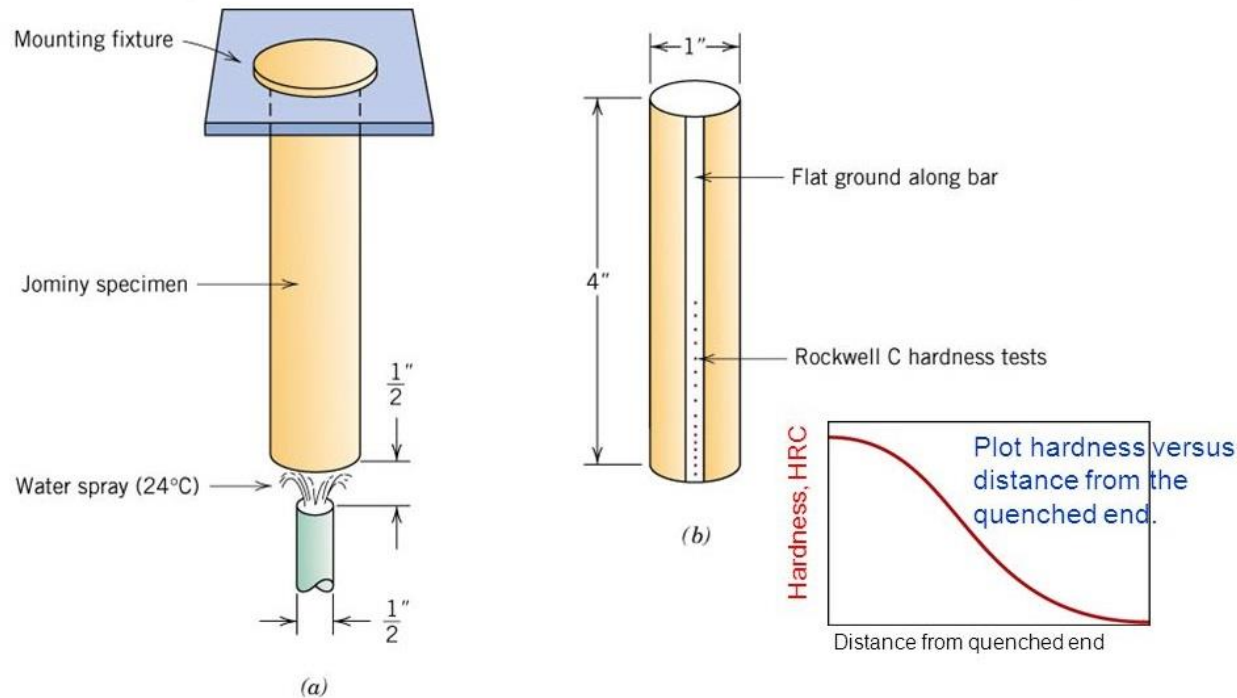
In **martempering**, steel is heated to above the critical range to make it all austenite. It's then quenched from the austenitising temperature into hot oil or molten salt at the approximate martensite start temperature (100°-200°C). The purpose is to delay the cooling for a long time to equalise the temperature throughout the piece.

Hardenability

The influence of alloy composition on the ability of a steel alloy to transform to martensite for a particular quenching treatment is related to a parameter called **hardenability**. For every steel alloy, there is a specific relationship between the mechanical properties and the cooling rate. *Hardenability* is a term used to describe the ability of an alloy to be hardened by the formation of martensite as a result of a given heat treatment. Hardenability is not “hardness,” which is the resistance to indentation; rather, hardenability is a qualitative measure of the rate at which hardness drops off with distance into the interior of a specimen as a result of diminished martensite content. A steel alloy that has a high hardenability is one that hardens or forms martensite, not only at the surface but also to a large degree throughout the entire interior.

The Jominy End-Quench Test

Hardenability -- Steels



A standard test to determine the hardenability is the Jominy end-quench test, in which a 1 in. diameter bar is austenised and the lower end quenched by a jet of water at a specific flow rate and temperature. The cooling rate is thus maximum at the end sprayed by water and decreases along the length of the bar.

Surface Treatments

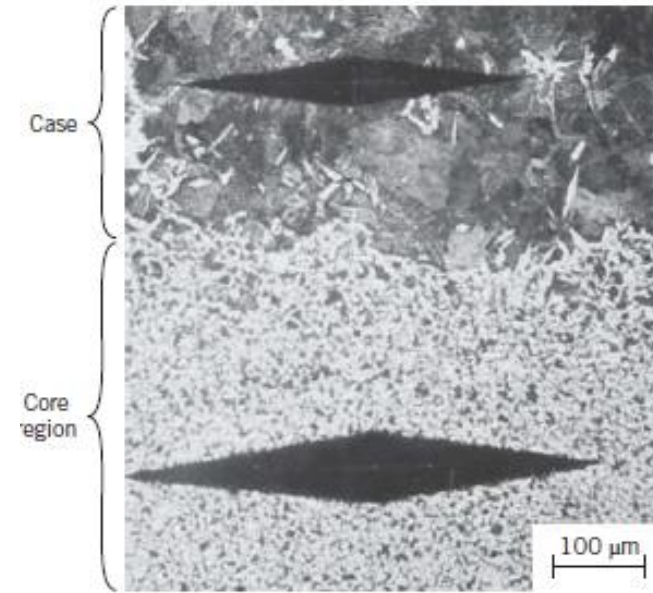
During machining operations, small scratches and grooves are invariably introduced into the workpiece surface by cutting-tool action. These surface markings can limit the fatigue life. It has been observed that improving the surface finish by polishing enhances fatigue life significantly.

Case hardening is a technique by which both surface hardness and fatigue life are enhanced for steel alloys. This is accomplished by a **carburizing or nitriding** process by which a component is exposed to a carbonaceous or nitrogenous atmosphere at elevated temperatures. A carbon- or nitrogen-rich outer surface layer (or *case*) is introduced by atomic diffusion from the gaseous phase. The case is normally on the order of 1 mm deep and is harder than the inner core of the material. A carbon-rich outer case may be observed for the gear shown in the figure; it appears as a dark outer rim within the sectioned segment.

Surface Treatments



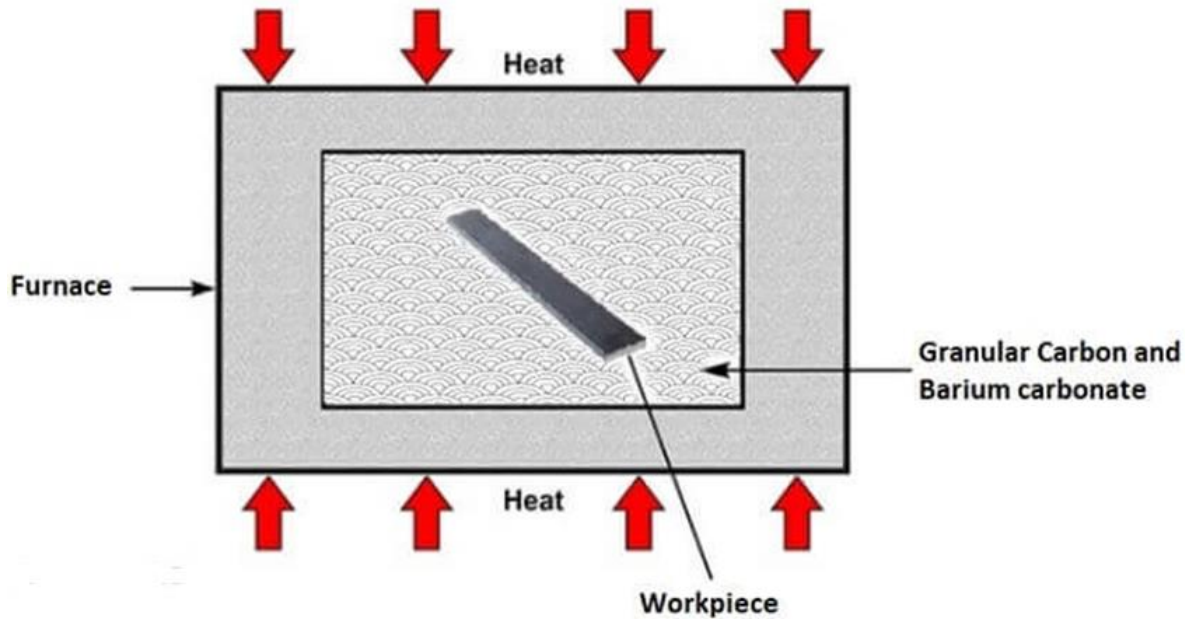
The steel gear shown above is case hardened, that is, its outer surface layer was selectively hardened by a high-temperature heat treatment during which carbon from the surrounding atmosphere diffused into the surface.



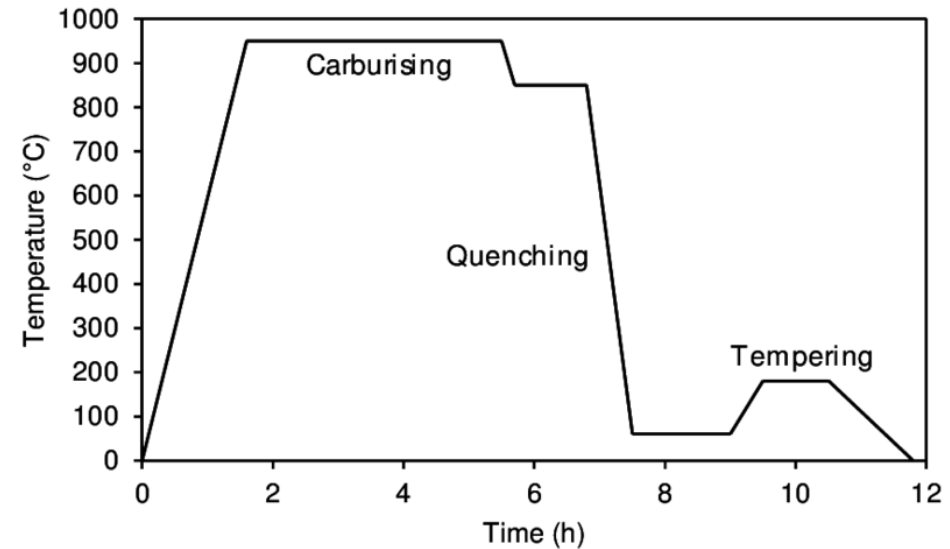
Photomicrograph showing both core (bottom) and carburised outer case (top) regions of case-hardened steel. The case is harder, as attested by the smaller microhardness indentation. 100×

Carburising

In case hardening, carbon is added to the surface of various alloys, giving the material a hard, wear-resistant outer layer while preserving a softer, more ductile core. The process is accomplished by heating the metal in a carbon-rich atmosphere above transformation temperature for a pre-determined time. After carburising, parts are quenched to harden the surface carburising layer.



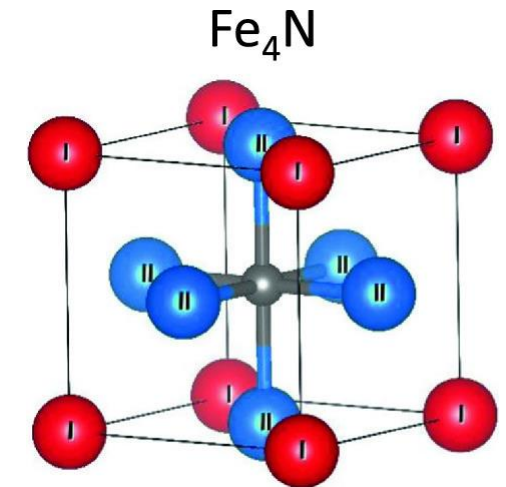
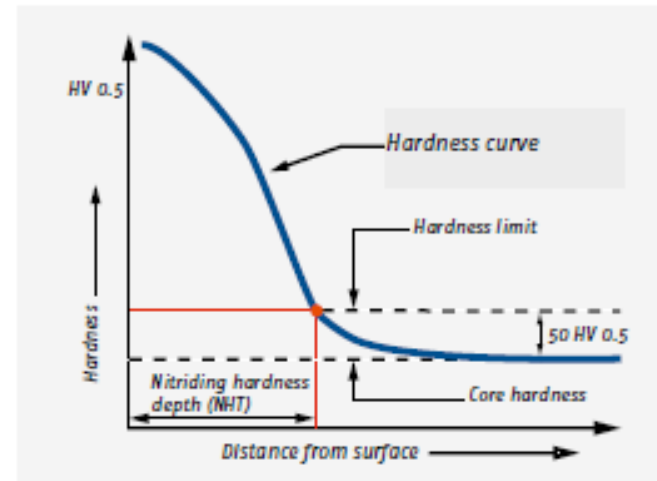
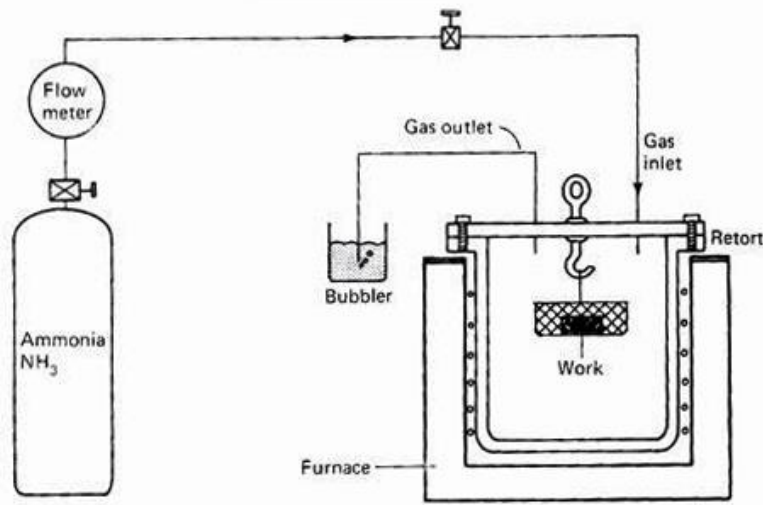
Carburising



Heat treatment after carburising

Nitriding

Nitriding is a **heat-treating process that diffuses nitrogen into the surface of a metal to create a case-hardened surface**. The added nitrogen combines with iron and other alloying elements in the metal's composition to form hard metallic nitrides.

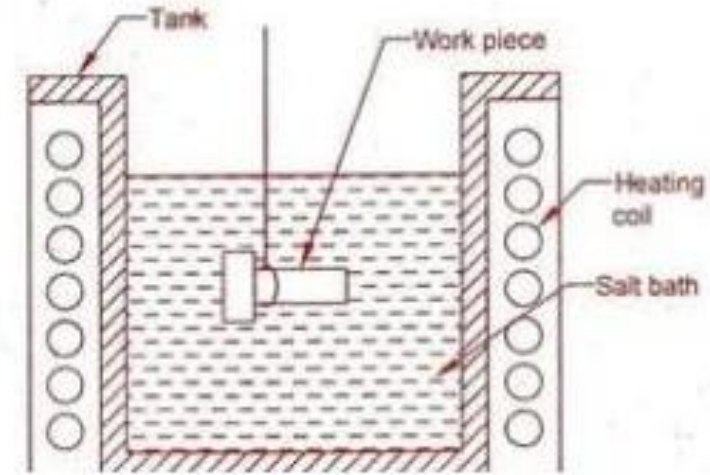


Cyaniding

A process in which an iron-base alloy is heated in contact with a cyanide salt so that the surface absorbs carbon and nitrogen. Cyaniding is followed by quenching and tempering to produce a case with a desired combination of hardness and toughness.

Cyaniding is a process where steel components are heated in the presence of **sodium cyanide or other cyanide salts**. When the metal is exposed to temperatures typically ranging from 760°C to 900°C in this environment, both carbon and nitrogen are assimilated onto the surface. This forms a case that is both hard and wear-resistant, with depths generally not exceeding 0.25 mm.

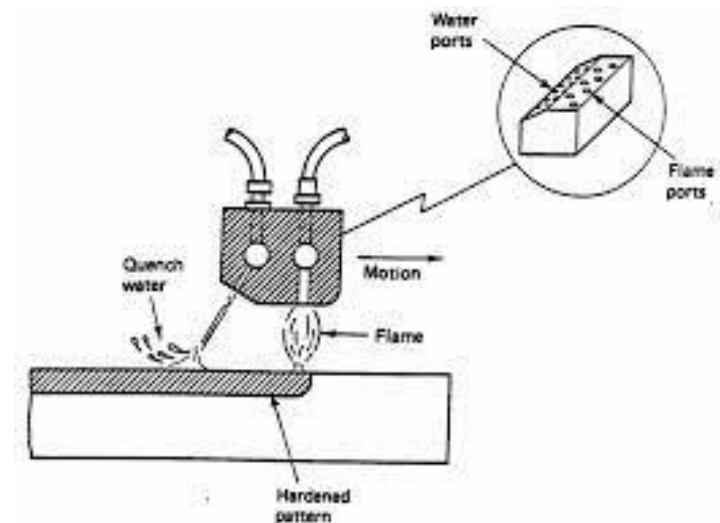
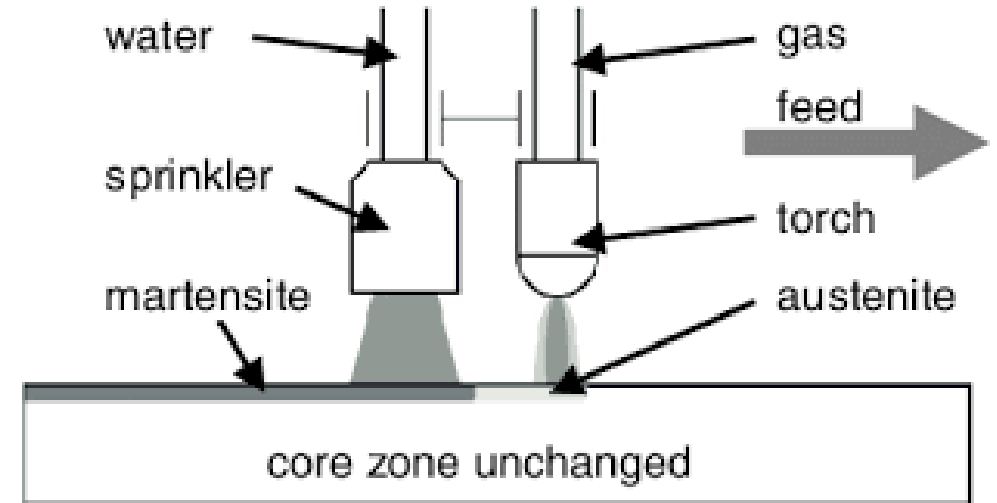
CYANIDING



Cyaniding vs Carbonitriding		
More Information Online WWW.DIFFERENCEBETWEEN.COM		
	Cyaniding	Carbonitriding
DEFINITION	Cyaniding is a type of case hardening process in which sodium cyanide is used	Carbonitriding is a type of case hardening in which a gaseous atmosphere is used for the hardening process
SURROUNDING	A bath of sodium cyanide	A gaseous atmosphere consisting of ammonia and hydrocarbon gases
TEMPERATURE	Around 871 to 954 Celsius degrees	For quenching the object, the temperature should be around 445 to 885 C° and if the metal surface is not quenched, the temperature would be around 649 to 788 C°

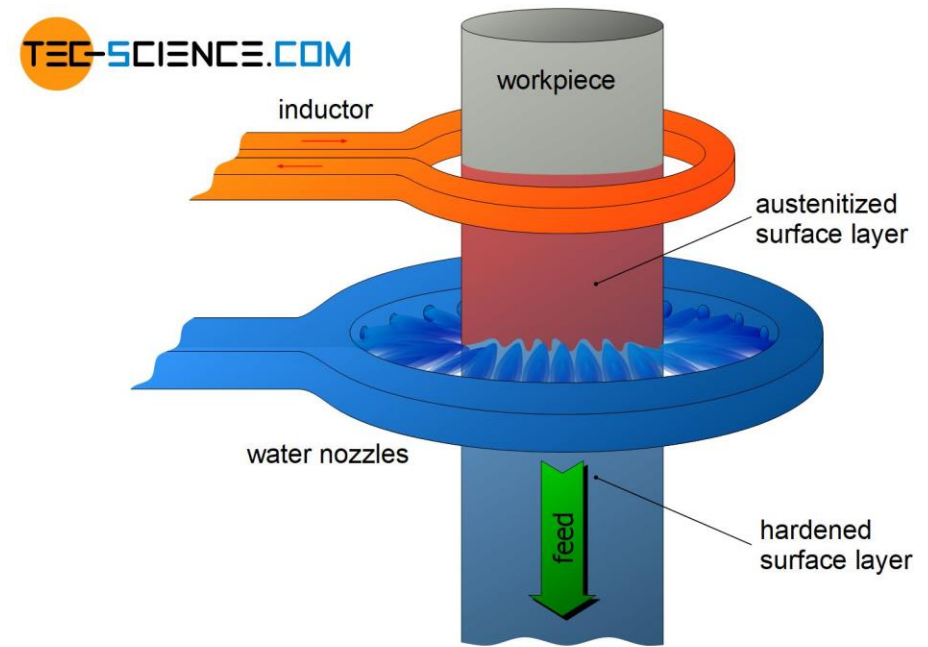
Flame hardening

Flame hardening is a surface hardening process to produce a hard wear-resistant surface on the applied part. It is generally used on medium carbon, mild or alloy steels or cast irons. This hardening process involves rapid heating of metal and then subsequently cooling it. The aim is to harden the metal while maintaining malleability for subsequent working. It requires the direct impingement of a high-temperature flame from a well-positioned burner with quenching capabilities. This hardening process is often used on the tooth flank of a gear wheel and also on the surface of a rail. The key parameter in flame hardening is achieving a very high and controlled temperature. As a fuel gas, Acetylene (ethyne) has proven to have the best heating properties for this process.



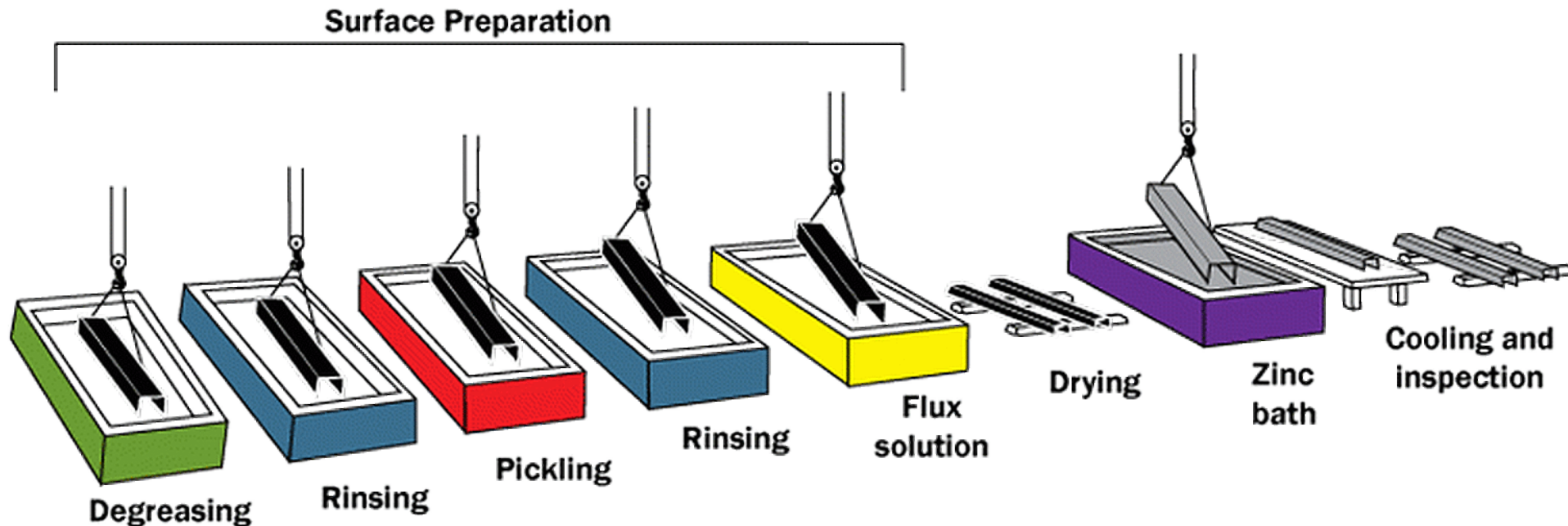
Induction hardening

Induction hardening is a method of quickly and selectively hardening the surface of a metal part. A copper coil carrying a significant level of alternating current is placed near (not touching) the part. Heat is generated at, and near the surface by eddy current and hysteresis losses. Quench, usually water-based with an addition such as a polymer, is directed at the part or it is submerged. This transforms the structure into martensite, which is much harder than the prior structure.



Metal coating-hot dipping

Hot-dip galvanising (HDG) is a **process that coats fabricated steel with zinc by immersing it in a bath of molten zinc**. The zinc coats the surface of the base metal, forming a zinc-iron alloy layer. This process helps protect the metal from corrosion.



Process of Hot Dip Galvanizing

Step 1 – Degreasing : It is the first stage, where there is a need to separate the grease and oil. It is done with the help of a hot alkaline cleaning solution.

Step 2 – Rinsing : The second stage of hot dip galvanizing process is rinsing. It is essential as it helps remove any alkaline present on the metal or steel surface. Also, to avoid any delay or contact alkaline with a pickling solution, there is a need to properly rinsing to it.

Step 3 – Scale Removal or Pickling : The third stage of hot dip galvanizing helps remove all the dust, surface rust, and mill scale from the metal. The cleaning process is done by putting the steel material in an acid bath.

Step 4 – Rising After Pickling : Once the pickling process is completed, there is a need to get steel appropriately rinsed. It is the stage that help to remove residual acid or iron salt in the metal. Once the rinsing process is complete, the steel is ready for further processing.

Step 5 – Pre-Fluxing : Before the steel is dip in molten zinc, there is a need to get the surface clean from the material. This steel is clean with the help of dipping the metal into a hot flux solution. It contains zinc ammonium chloride.

Step 6 – Zinc Bath : The metal must undergo a zinc bath to form a protective layer with the steel and It is an important stage of the hot dip galvanizing process.

Step 7 – Finishing : The final process where there is a need to have a finish product delivered to the team. Hence, following this stage will help complete the work effectively and conveniently in galvanizing work.